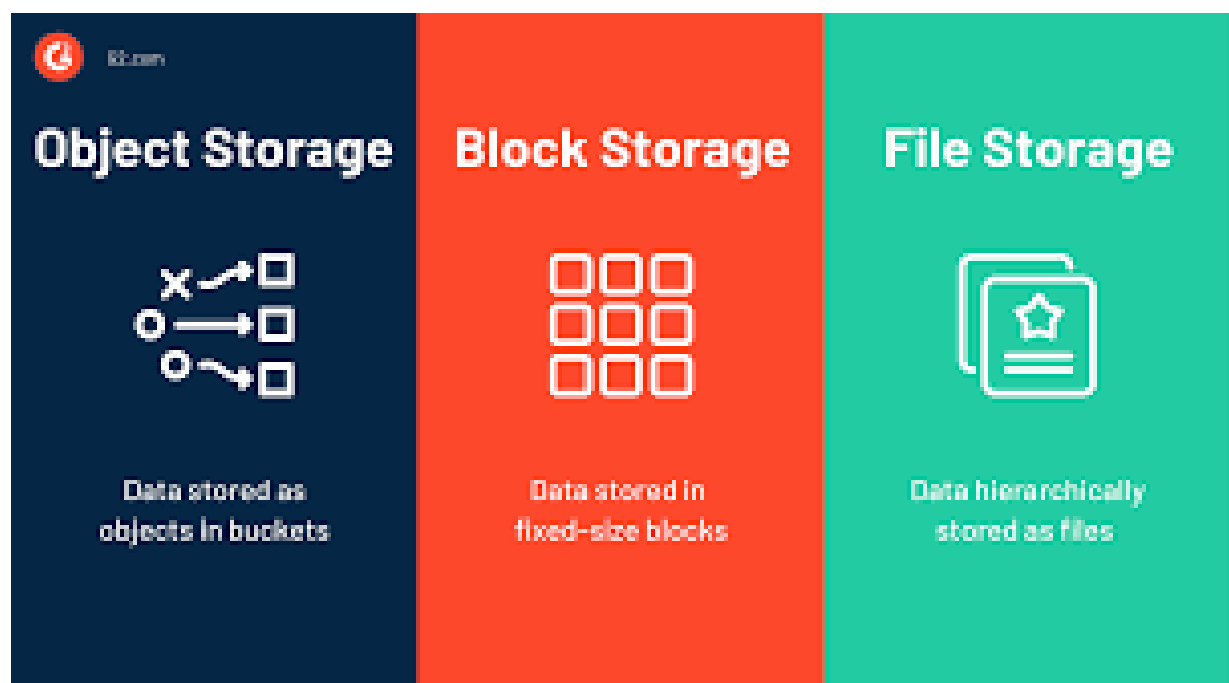


Block, Object, and File Storage

Storage is an important component because it determines how data is stored, accessed, and managed. Different storage types are used depending on application requirements and data structure.

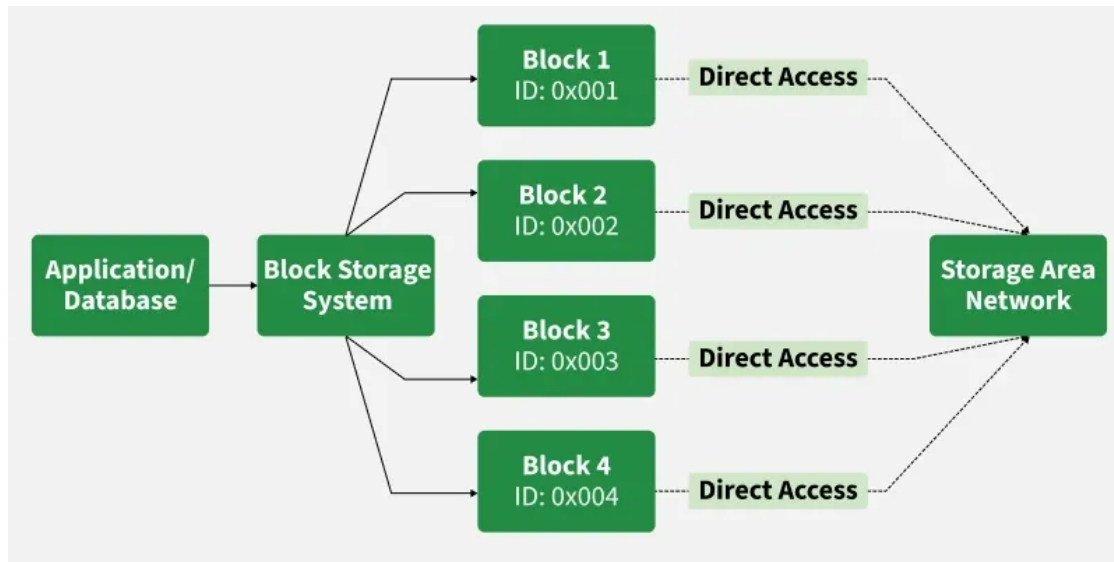
- Block Storage stores data in fixed-size blocks and is commonly used in databases and virtual machines.
- File Storage organizes data in files and directories, making it suitable for shared file systems and document storage.
- Object Storage stores data as objects with metadata and unique identifiers, ideal for large-scale unstructured data.



1. Block storage

Block storage is a technique for keeping data in chunks or blocks of an exact size. Every block has its own address, functions independently, and can store any kind of data. Block storage lacks a predetermined structure, unlike file storage, which arranges data in a hierarchy. It is

frequently utilised in systems like databases and virtual machines that require great performance and scalability.



Block Storage Architecture

Block storage doesn't inherently "understand" the data it stores - it only stores raw bytes. The file system on top of block storage interprets it into files. So block storage itself isn't aware of formats or data types.

Example

Consider a cloud-based database service where you need to store a large amount of structured data. The data is broken into smaller pieces (blocks) and distributed across a storage area network. When you access the database,

- The system retrieves the required blocks and reassembles them into meaningful data for your application.
- Amazon Elastic Block Store (EBS) is a real-world example of block storage.

Working

Imagine a warehouse where items aren't organized by category or aisle, but simply placed in numbered bins. Each bin (block) can hold anything, and you access items directly by their bin number. This direct-access model eliminates the overhead of navigating folder hierarchies, resulting in exceptional speed.

Features

Block storage stores data in fixed-sized blocks, making it suitable for high-performance and low-latency applications.

- **High Performance:** Block storage is perfect for high-performance applications since it is designed for quick read/write operations.
- **Flexibility:** Since it does not impose a particular structure, it allows data to be stored in any format.
- **Scalability:** Blocks can be added or removed easily to scale the storage up or down.
- **Independence:** Each block operates independently, enabling precise control and management of data.
- **Use in Distributed Systems:** Block storage can be distributed across multiple servers for redundancy and improved performance.

2. Object Storage

With object storage, data is kept as discrete units known as "objects." A unique identity, metadata (information about the data), and the actual data are all contained in each item. Object storage is hence very flexible, scalable, and appropriate for storing vast amounts of unstructured data, such as backups, videos, and pictures.

- Object storage doesn't use fixed-sized blocks or a hierarchical file system like file or block storage does.
- Instead, it organizes data into a flat structure, which is easier to scale and manage in distributed environments.

Note: Object storage is ideal for write-once, read-many (WORM) workloads, not for frequently modified data.

Understanding the Object Model

Think of object storage as a vast library where every book has a unique ISBN, a detailed catalog card, and can be retrieved directly without knowing which shelf it's on. The flat structure eliminates hierarchical limitations, allowing the system to scale horizontally across unlimited storage nodes.

Example

Object storage is widely used for storing large amounts of unstructured data in cloud-based applications.

- **Media Streaming:** Netflix and YouTube store billions of video files as objects, serving them to millions of concurrent viewers.
- **Backup and Archival:** Companies use object storage for long-term data retention, leveraging its durability and low cost per gigabyte.
- **Big Data Analytics:** Data lakes built on object storage allow analysts to store raw data in any format and process it at scale.
- **Static Website Hosting:** Web assets like images, CSS, and JavaScript files are served directly from object storage.

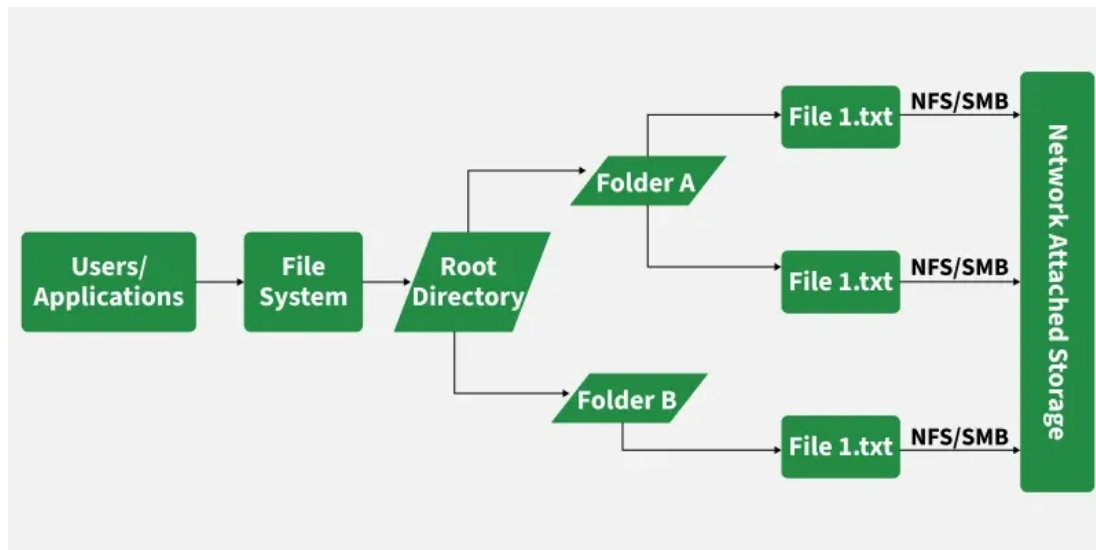
Features

Object storage systems are designed to handle massive volumes of unstructured data efficiently.

- **Scalability:** Object storage is perfect for cloud applications because it can manage massive volumes of data.
- **Metadata Richness:** Metadata is stored in every object to help with data management, indexing, and searching.
- **Global Accessibility:** Objects can be accessed via HTTP/HTTPS, making it suitable for web-based applications.
- **Cost-Effective for Unstructured Data:** Large amounts of unstructured data, such as logs, media files, and backups, are ideal for this storage.
- **Resilient and Durable:** To provide stability and fault tolerance, object storage systems frequently duplicate data across different locations.

3. File Storage

Similar to how we arrange files on a computer, file storage is a conventional technique of storing data in a hierarchical system of files and folders. Every file has a name and directory path, which helps access and navigation. Applications that need regular updates and organized data management are best suited for it.



File Storage in System Design

The Hierarchical Model

File storage organizes data in a tree structure with directories (folders) containing files and subdirectories. Each file has a path (like `/projects/2024/report.pdf`) that defines its location, making navigation logical and straightforward.

Example

Amazon EFS, Google Filestore, or Azure Files for cloud file storage analogs

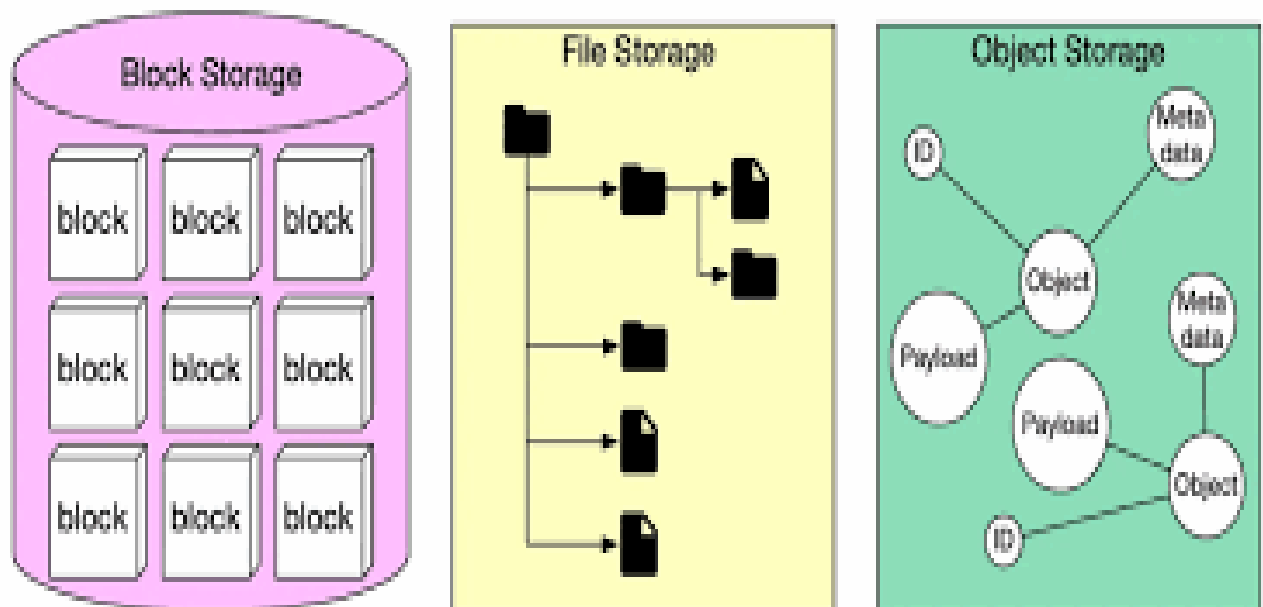
- **Corporate File Shares:** Departments share documents, spreadsheets, and presentations through networked drives with folder hierarchies mirroring organisational structure.
- **Content Management:** Web servers host files in organised directories, serving websites through familiar path-based URLs.
- **Development Environments:** Developers work with codebases organised in folder structures, relying on file storage for source control integration.
- **Home Directories:** User workspaces on servers maintain personal files in individual, permission-protected directories.

Features

File storage systems organize and manage data in a traditional file-and-folder structure, making them easy to understand and use.

- **Hierarchical Organization:** Data is stored in a clear folder-and-file structure, making it easy to locate and manage.
- **Simplicity:** File storage systems are easy to set up and use for small-scale applications.
- **Compatibility:** Works well with legacy applications and systems that require traditional file access methods.
- **Shared Access:** Supports multi-user environments with file permissions and version control.
- **Data Integrity:** Ensures consistency and integrity through locking mechanisms during file updates but distributed file systems (like NFS) can experience locking issues and performance bottlenecks under concurrency.

Block Storage Vs Object Storage Vs File Storage



Differences between them are as follows:

Block Storage	Object Storage	File Storage
Divides data into fixed-size blocks, each with a unique identifier.	Stores data as objects with metadata and a unique ID in a flat structure.	Organizes data in a hierarchical structure of files and folders.
Ideal for databases, virtual machines, and transactional workloads requiring high performance.	Best for storing large amounts of unstructured data, like multimedia files or backups.	Suitable for structured file storage and shared file access, such as documents and spreadsheets.
High performance and low latency, especially for read/write operations.	Optimised for scalability and durability, not real-time performance.	Moderate performance; dependent on file system and storage device.
Scales well but may require manual configuration for capacity expansion.	Highly scalable; can handle massive amounts of data across distributed systems.	Limited scalability compared to object storage; suitable for smaller systems.
Minimal metadata, often handled by the application layer.	Extensive metadata stored with each object, enabling advanced search and analytics.	Basic metadata, such as file name, type, and permissions.

Requires manual backup or snapshot configurations for data durability.	Highly durable with built-in redundancy across multiple locations.	Data durability depends on the underlying file system and backup strategies.
AWS EBS, Google Persistent Disks, SAN (Storage Area Network).	AWS S3, Azure Blob Storage, Google Cloud Storage.	Network Attached Storage (NAS), Shared Drives, Local File Systems.

Storage tiers IN CLOUD COMPUTING

Tiered storage is a system or method for assigning data to various types of storage media based on a range of requirements for cost, availability, performance, and recovery. Several factors determine which type of media, such as cloud storage, solid state storage arrays, disk or tape, is ideal for storing a particular category of data. These factors include cost, data recoverability, and data availability requirements.

Tiered data storage is part of the foundation of information lifecycle management (ILM) and helps companies reduce total storage costs while ensuring compliance and performance. For example, data retained for regulatory needs can be archived to save money, while data needed for restoration in the event of corruption, data loss, failed data migration, or other critical failure can speed recovery if it is stored locally.

WHAT IS TIERED STORAGE?



- ALLOCATING DIFFERENT CATEGORIES OF DATA TO VARIOUS TYPES OF STORAGE MEDIA IN ORDER TO MINIMIZE THE TOTAL STORAGE COST.
- TIERS ARE DETERMINED BASED ON BELOW GIVEN PARAMETERS;

Cost

Performance

How often it
is accessed

What are the Data Classes for Tiered Storage?

Typical tiered storage data classes include:

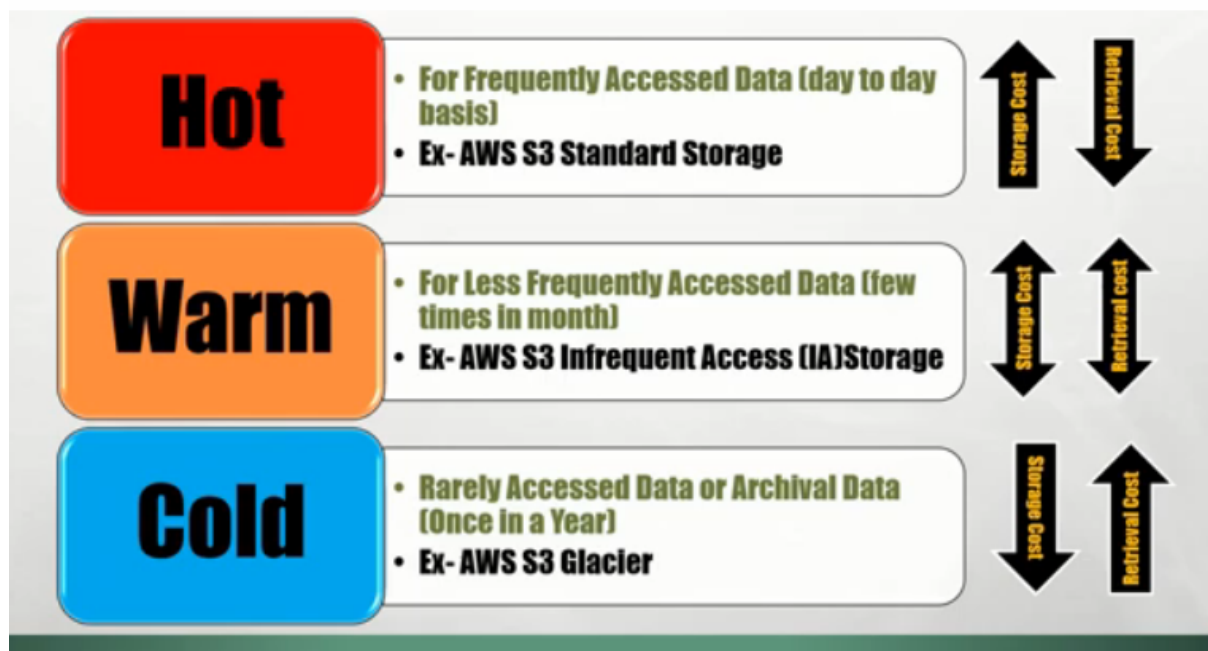
Mission critical data. This data must always be stored at the highest tier because it supports high-speed applications such as customer transactions. Inability to access this kind of mission critical data means poor performance, lost business, delays, and other problems that negatively impact profitability.

Hot data. This data class is in constant use, so it demands a relatively high level of tiered storage. Data like this runs email, and applications such as ERP or CRM—anything the organization's everyday operations depend on. At this tier performance is still very important, but cost is also a factor.

Warm data. This data is typically a few days old, and includes data on completed transactions and emails from days past—things that are used less frequently, but still must remain readily accessible. At this level, cost becomes the more important consideration, subject to a threshold for minimum performance.

Cold data. This sort of data has to be retained, but it may never be accessed again. Perhaps there are regulatory reasons to keep it, or it might be unstructured data that has big data mining uses in the future. Cold data like this should be stored based on low cost, with access times of minutes or hours.

Except for very small businesses, manual storage tiering is almost always too impractical and time consuming to work well. Most tiered storage systems deploy automated storage tiering, moving data through the tiers automatically as it moves through its lifecycle, cooling. This kind of storage tier data management system software may be a standalone solution, part of a cloud storage gateway, or part of a single storage system.



What is Multi-Tiered Storage Architecture?

In a multi-tiered storage architecture, the various storage media are organized hierarchically. The highest performance storage media is Tier 0 or Tier 1, with the other tiers, Tier 2, Tier 3, and so on, following. Today it is common to see tiered storage models with five or more tiered storage levels. That said, some experts anticipate fewer tiers, with the most valuable data being stored on flash arrays and less-valuable data being stored in the cloud. The common factor is that a well-designed tiered storage system will ensure that each tier has a different, useful blend of performance, cost, and capacity.

Typically, flash storage or 3D Xpoint-based solid state drives comprise Tier 0 or Tier 1. Tape storage systems, high performance SAS or fiber channel drives, RAID arrays containing those kinds of drives, optical

disks, lower performance SATA drives, and cloud-based storage systems may all be part of lower tiered storage levels.

Tiered Storage vs Hierarchical Storage Management

Although tiered storage and hierarchical storage management (HSM) are sometimes confused, they are not exactly the same. HSM typically refers to a data movement system in which movement happens automatically to some kinds of media depending on how often they are used. Tiered storage is where an HSM system might store its data.

Advantages of Tiered Storage

The advantages of data tiering storage include:

Reduced storage costs. Tiered storage allows companies to store each class of data based on the minimum performance that it requires and the lowest cost storage that can handle those requirements. This in turn eliminates the problem of paying for unneeded high performance storage.

Increased storage efficiency. High performance disk storage systems are often inefficient. Tiered storage systems move less critical data to lower tiers, reducing demand on short-stroked and RAID storage and improving storage efficiency.

Reusable storage equipment. Equipment that doesn't perform well enough for Tier 1 can still be reused in a tiered storage system.

Improved disaster recovery. Tiered storage infrastructures can significantly reduce the costs associated with achieving a zero recovery time objective (RTO) and recovery point objective (RPO) by reducing the amount of data the DR system must recover during a disaster. RPO and RTO are among the most important parameters of a data protection or disaster recovery plan because classifying data correctly ensures the data is in the best possible location for recovery at all times.